

FDE

v0.6 beta

v0.6-beta

2026-05-24 freeze

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+

README.md

· PLAN.md

· docs/deployment.md

1.

2.

3.

4. — 10 sidebar

5.

6.

7.

8. FAQ

9.

1.

· FDE Field Delivery Engineer

"

vendor"

"

+

+

"

dev

admin-webhttp://localhost:5173admin / lead / pm / finance / engineer / vendor 6+ +

webadmin-http://localhost:5173engineer/ /

cockpit-screenhttp://localhost:51745 Tab

seed_demo

adminadmin123lead

financefin123finance

pm1pm123pm

eng1eng123engineer

v1	v4	demo123	vendor	vendor
X-Cockpit-Token: cockpit-dev-token				

2.

6

admin	—
lead	—
pm	—
finance	—
engineer	User.engineer_id → engineers
vendor	Vendor vendor User.vendor_id → vendors

2.1

	admin	lead	pm	finance	engineer	vendor
	✓	✓	✓	✓	✓	✗ /expense
	✓	✓	✓	✗	✗	✗
	✓	✓	✓	✗	✗	✗
	✓	✓	✗	✓	✗	✗
+	✓	✓	✓	✓	✗	✗
vendor	—	—	—	—	—	✓
	✓	✓	✓	✗	✗	✗
FDE	✓	✓	✓	✓	✓	✗
	✓	✓	✓	✓	✗	✗

	admin	lead	pm	finance	engineer	vendor
	✓	✓	✓	✗	✗	✗
.	✓	✗	✗	✗	✗	✗
/ /	—	—	—	—	✓	✗

2.2

monthly_cost_to_telecom	admin / lead / finance
+	admin / lead
	admin / lead / finance
	admin / lead / finance
confidential	admin / lead / pm / finance

2.3

/ no_revenue	admin / lead
SalesTransferLog	admin / lead
/ / vendor	admin / lead
	engineer / vendor vendor / pm
approve / reject / mark paid	admin / lead / finance
vendor + vendor_id	admin

3.



freeze

[README §1.7](#)

3.14

4

	finance
gross_amount	project_revenues.gross_amount

non_service_expense / project_revenues.non_service_expense				finance
/				
amount	team_revenue	project_revenues.amount		finance
				sales
outsource_benchmark_amount projects.outsource_benchmark_amount				
				vendor

- non_service_expense ≈ 70% × gross_amount
- team_revenue ≈ 20% × gross_amount
- ~10%

3.2

A		admin / lead / finance	VSF - Σ 全部支出 (含外包工程师支出)	—
B	-	admin / lead / finance	团队入账 - VSF / 0 - 外包估算 no_revenue	—
C		+	外包估算 - 团队入账 won / 外包估算 no_revenue	revenue/cost/margin A/B

 CI response

tests/test_cockpit_isolation.py /api/cockpit/* revenue / cost / margin / team_margin / vendor_fees

JSON

3.3 C

backend/app/services/profit.py

compute_cockpit_savings_and_value()

PROJECT_KIND_REVENUE

bid_outcome == 'won'

team revenue

savings = Σ (outsource_benchmark_amount - Σ project_revenues.amount)
per project where bid_outcome='won'

PROJECT_KIND_NO_REVENUE

status ∈ {closing, archived}

/

/ cancelled

value_created = Σ outsource_benchmark_amount per project where status in {closing, archived}

total_c_view = savings + value_created

bid_outcome

C

		savings
pending	/	✗
won		✓
lost		✗
escaped		✗

1. sales 拍报价 + 立项

→ bid_outcome=pending, 驾驶舱不动
2. 投标中 ...

→ 不动
3. 客户告知中标, sales 改 bid_outcome=won

→ 驾驶舱立即加 (benchmark - team_revenue)
4. 财务录入 ProjectRevenue.amount

→ 驾驶舱按新公式重算
5. 项目正常交付 closing / archived

→ 驾驶舱不变 (已计入)
6. 客户违约跑单, 改 bid_outcome=escaped

→ 驾驶舱立即扣除

v0.6

savings

v0.4

backend/app/services/profit.py compute_company_margin_lift()

```
老外包模式毛利率 = (gross - benchmark - non_service_expense) / gross
                    # 没有 FDE 团队时，公司直接付外部 benchmark + non_service

FDE 模式毛利率 = (gross - VSF - non_service_expense) / gross
                 # 团队自管，公司付 VSF (含团队真实利润) + non_service

利润率提升 = FDE 毛利率 - 老外包毛利率    # 单位：百分点
```

seed	6.34%	FDE 9.54%	+3.20
------	-------	-----------	-------

4.

10	sidebar	§2.1
----	---------	------

4.1 /dashboard

4		
1.		
2.	lead / finance / admin	
3.	VSF	
4.	5	
vendor / engineer		KPI

4.2 /project

5	Tab	/	/	/	/
		kind	revenue/no_revenue	status	bid_outcome
		NeedParty	/		logo
		SalesPerson			
		ProjectRevenue	finance	+ gross_amount + non_service_expense	
		RenewalAttempt		pending/won/lost + 6	
		Excel		+	

4.3 /engineer

4	Tab				
			Assignment	/	/
				admin/pm	
		engineers	CRUD	+	
		Skill	—	+	+ L1/L2/L3

Skill

4.4 /profit admin / lead / finance

4 Tab

- A KPI = VSF – + vs FDE
- B
- B
- C cockpit screen

 — lost pre-sales VSF tooltip

4.5 /expense

3 Tab vendor 1

- **Vendor** 8 / / / / / / / /
pending → approved/rejected → paid
- **Vendor** VSF vendor
- **Supplier**

vendor

- vendor
- approve/reject/paid
- tab

4.6 /efficiency

KPI 4 +

- / / 14 /
 -
- = admin ↔ / /

4.7 FDE /knowledge

- KnowledgeAsset CRUD + AssetReference
- 7 / / / / / /
 - public / internal engineer / confidential
 - engineer
 - X N → " "

4.8 /capability

3 Tab

- EngineerSkillSnapshot SVG admin
- TrainingRecord CRUD lead/finance
- IDP IDP CRUD / /

4.9 /relationship

2 Tab

- ProjectRetrospective / / / 1-5
- /
- §4.2 " "

4.10 · /users admin

CRUD

- + +
- engineer engineer_id engineers
- vendor vendor_id vendor
- / /

5.

engineer 4 / / /
engineer /dashboard

5.1 /my-assignments

Assignment

- admin/pm /
- PM
-

5.2 /my-timesheets

Timesheet + + + §7.3

- pending pm/admin
-

5.3 /my-expenses

ExpenseRequest §7.4

6.

http://localhost:5174

60

01	KPI	+ HK 18		+	logo	+	
02		vs FDE			vendor		
03	/	/ 14	/	+		+ 5	KPI
04		/	/	/	+	+	
05			+	+	+	+ Top	+ 6x3

Tab	/api/cockpit/*	CI	A/B
-----	----------------	----	-----

- `--cockpit-accent #00e5ff`
- `--cockpit-accent-2 #7c4dff`
- `brag --cockpit-accent-3 #ff4081`
- `brag-2 --cockpit-accent-gold #ffe082` /
- `growth --cockpit-accent-green #67ff8a` delta

7.

7.1 → →

1. sales 在 /project 新增项目，填外包估算（必须真实询价依据）
→ kind=revenue（默认），bid_outcome=pending
2. 投标过程中，sales 可以改报价（外包估算）
3. 客户中标通知到达：
sales 把 bid_outcome 改 won → 驾驶室降本数字立即跳升
4. PM 在 /engineer 派单（Assignment），工程师在工程师端「接单」
5. 工程师在 /engineer 端录工时（按周或按月）
PM/admin 审批工时
6. 项目过程中产生外部支出：
工程师 / vendor 提交 expense → finance/lead 审批 → 付款
7. 客户分期付款时，finance 在 /project 收入 Tab 录入 ProjectRevenue
含 gross_amount / amount(team_revenue) / non_service_expense
8. 项目交付完成：sales 改 status 为 closing / archived

9. PM/lead 在 /relationship 写复盘报告（含满意度）
10. 客户考虑续单：在 /project 续单 Tab 创建 RenewalAttempt 跟踪

7.2

PM 创建 Assignment (指定工程师 + 项目 + 计划工时 + 起止日期)
↓ status = 'proposed'
工程师收到推送 (站内信 / 飞书, Phase 4)
↓
工程师在「我的派单」里：
├─ 接受 → status = 'active', 开始记工时
└─ 拒绝 → 必须填原因, 状态回到 PM 看, PM 改派或撤销

project 完成时：
PM 把 Assignment 改 status = 'ended'

7.3

	09:00-18:00	1.0×
	18:00	1.5×
/ /		1.5×

=

7.4

提交 (engineer / vendor / pm)
→ status = 'pending'
↓
审批 (admin / lead / finance)
├─ approve → status = 'approved'
└─ reject (带 reject_reason) → status = 'rejected'
 ↓
 财务付款标记
 → status = 'paid'
 不进 cost 累计

vendor

approve/reject/paid

vendor

7.5

项目 closing / archived 后 :

PM 或 lead 在 /relationship 创建 ProjectRetrospective

- 做对的事
- 要改的事
- 行动项列表 (每项有责任人 + ddl)
- 客户满意度 1-5 星
- 闭环开关 : 行动项全部跟到完成时勾上

驾驶舱 Tab 5 (项目进度视图) 按"闭环率"算团队执行力

8. FAQ

Q	engineers	A	admin	/users		/engineer
			User	engineer_id	vendor	
	vendor_id					
Q	escape	A	/project	bid_outcome	escaped	
	savings		VSF			
Q		A	/project →	Tab	ProjectRevenue	
	gross_amount		non_service_expense			
Q		A	training			
	/capability →	Tab	IDP			
Q	vendor	A	vendor			
	+ vendor +		/ vendor			
Q	L1/L2/L3	A	L1=	/ L2=	/ L3=	
	CCNA=L1 CCNP=L2 CCIE=L3		x			
Q		A	Tab	60	fetch	
Q		A	admin	/users	Phase 4	
Q		A	/engineer	→	+	,
Q		A	/profit →	Tab		
	revenue / cost / margin	B	– VSF			

9.

9.1

```
docker compose -f docker-compose.local.yml up --build
```

[docker-compose.local.yml](#)

9.2 /

```
# 本机 export
docker save manpower-management-platform-{backend,admin-web,cockpit-screen} \
  postgres:16-alpine | gzip > manpower-images.tar.gz

# 目标机
gunzip -c manpower-images.tar.gz | docker load
docker compose -f docker-compose.offline.yml up -d
```

[docker-compose.offline.yml](#)

9.3 ()

[docs/deployment_ctyun.md](#) [docker-compose.prod.yml](#)

- `.env.production` `JWT_SECRET` / `FIELD_ENCRYPTION_KEY` / `POSTGRES_PASSWORD` / `DEFAULT_ADMIN_PASSWORD` / `COCKPIT_TOKEN`
- `COCKPIT_ALLOWED_IPS` `IP` `token`
- `prod compose` `pg-backup service` `pg_dump` `30`
- `HTTPS` `nginx` `nginx/certs/` `Let's Encrypt`

9.4

```
# 备份
docker exec manpower-pg pg_dump -U manpower -d manpower -Fc > backup-$(date +%F).dump

# 恢复
docker exec -i manpower-pg pg_restore -U manpower -d manpower -c < backup-2026-05-24.dump
```

9.5

1. `git pull`
2. `docker compose -f docker-compose.prod.yml build`
3. `docker compose -f docker-compose.prod.yml up -d`
4. `alembic upgrade head`

5. migration apply docker compose logs backend | grep alembic

9.6

DATABASE_URL	postgres
JWT_SECRET	JWT
FIELD_ENCRYPTION_KEY	
DEFAULT_ADMIN_USERNAME/PASSWORD	admin
COCKPIT_ALLOWED_IPS	IP
COCKPIT_TOKEN	token
CORS_ORIGINS	
UPLOAD_DIR	

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- → [README §1.7](#) §3
- → §4 §2.1
- → §2 §4
- Tab → §6 [CockpitFrame.vue](#)